Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Martins View Jersey Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
28336 W Bambauer RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1941

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0054-0110-0023-0000 0063-0050-0028-0000

B. OPERATORS

Martins View Jersey Dairy, Inc.

Operator name: Martins View Jersey Dairy, Inc.

Telephone no.: Landline: ; Cellular: (209) 678-1156

Mailing Address Number and Street	City	State	Zip Code
1000 Red Lion CT	Newman	CA	95360

This operator is responsible for paying permit fees.

C. OWNERS

Amarante, Pat

Legal owner name: Amarante, Pat

Telephone no.: Landline: (209) 854-3775; Cellular: (209) 605-1971

Mailing Address Number and Street	City	State	Zip Code
1307 Costa Del Sol	Shell Beach	CA	93449

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	55	80	0
Number under roof	0	0	398	0	0	0
Maximum number	0	0	405	60	90	0
Average number	0	0	398	55	80	0
Avg live weight (lbs)	0	0	775	350		

Predominant milk cow breed: Jersey

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 4,601.25 *tons per reporting period*

Total nitrogen from manure: 47,077.70 *lbs per reporting period*

After ammonia losses (30% loss applied): 32,954.39 *lbs per reporting period*

Total phosphorus from manure: 7,564.26 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,146,427 *gallons*

Total nitrogen generated: 15,028.68 *lbs*

Total phosphorus generated: 2,785.93 *lbs*

Total potassium generated: 42,998.75 *lbs*

Total salt generated: 191,516.91 *lbs*

	6,290,628	*gallons applied*
+	0	*gallons exported*
-	3,144,201	*gallons imported*
=	3,146,427	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
DW #1	Ground water
DW #2	Ground water
IW	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
01/24/2023	Process wastewater *Martins View Jerseys Dairy 1 WW*	360,155.00 *gal*	664.00	132.00	1,920.00		9,300
04/14/2023	Process wastewater *Martins View Jerseys Dairy 1 WW*	618,328.00 *gal*	731.00	63.50	1,500.00		6,520
06/07/2023	Process wastewater *Martins View Jerseys Dairy 1 WW*	1,315,901.00 *gal*	731.00	63.50	1,500.00		6,520
08/29/2023	Process wastewater *Martins View Jerseys Dairy 1 WW*	509,975.00 *gal*	470.00	50.70	969.00		5,560
09/24/2023	Process wastewater *Martins View Jerseys Dairy 1 WW*	339,842.00 *gal*	470.00	50.70	969.00		5,560

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	17,127.93	1,781.24	36,854.14	172,621.31
Total imports for all materials	17,127.93	1,781.24	36,854.14	172,621.31

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/01/2023	Separator solids	48.00 *ton*	Dry-weight	43.4		21,500.00	6,010.00	40,500.00		74.40

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	1,168.22	326.56	2,200.61	40,425.98
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	1,168.22	326.56	2,200.61	40,425.98

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	5	5	1	none	X054-X110-X023-XXXX
Field 2	15	15	2	process wastewater	X054-X110-X023-XXXX
Field 3	15	15	2	process wastewater	X054-X110-X023-XXXX
Field 4	22	22	2	process wastewater	X054-X110-X020-XXXX X054-X120-X013-XXXX
Field 5	36	36	2	process wastewater	X063-X050-X030-XXXX
Totals for areas that were used for application	88	88	8		
Totals for areas that were not used for application	5	5	1		
Land application area totals	93	93	9		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/01/2021: Pasture

Crop: Pasture Acres planted: 5 Plant date: 11/01/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	75.00 *ton*	As-is		71.4	5,200.00	800.00	6,900.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	31.50	256.50	0.00
Total actual harvest content	15.00	156.00	24.00	207.00	677.82

Field 2

Field name: Field 2

Field 2

11/03/2022: Oats, hay

Crop: Oats, hay Acres planted: 15 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	131.43 *ton*	As-is		5.9	10,500.00	1,810.00	15,300.00		8.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	240.00	39.00	198.00	0.00
Total actual harvest content	8.76	184.00	31.72	268.12	1,391.76

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	278.85 *ton*	As-is		64.2	4,370.00	530.00	4,390.00		3.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	18.59	162.48	19.71	163.22	412.62

Field 3

Field name: Field 3

11/03/2022: Oats, hay

Crop: Oats, hay Acres planted: 15 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	131.43 *ton*	As-is		5.9	10,500.00	1,810.00	15,300.00		8.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	240.00	39.00	198.00	0.00
Total actual harvest content	8.76	184.00	31.72	268.12	1,391.76

Field 3

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	278.85 *ton*	As-is		66.2	4,230.00	390.00	3,750.00		2.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	18.59	157.27	14.50	139.43	339.30

Field 4

Field name: Field 4

11/03/2022: Oats, hay

Crop: Oats, hay Acres planted: 22 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	192.76 *ton*	As-is		6.2	7,830.00	1,540.00	16,900.00		8.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	240.00	39.00	198.00	0.00
Total actual harvest content	8.76	137.21	26.99	296.15	1,374.15

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	408.98 *ton*	As-is		67.0	4,160.00	540.00	4,310.00		2.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	18.59	154.67	20.08	160.25	306.74

Field 5

Field name: Field 5

12/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 12/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/29/2023	707.61 *ton*	As-is		72.4	4,440.00	700.00	6,270.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	19.66	174.54	27.52	246.48	1,106.70

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	650.65 *ton*	As-is		62.4	5,040.00	450.00	5,720.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	18.07	182.18	16.27	206.76	598.02

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/01/2021: Pasture

Field name: Field 1

Crop: Pasture

Plant date: 11/01/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	121.82	456,191.00 *gal*
Application event totals		0.00	0.00	0.00	121.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	8.99	0.00	0.00	497.90	528,000.00 *gal*
Application event totals		8.99	0.00	0.00	497.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	261.05	977,553.00 *gal*
Application event totals		0.00	0.00	0.00	261.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	217.54	814,628.00 *gal*
Application event totals		0.00	0.00	0.00	217.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	95.72	358,436.00 *gal*
Application event totals		0.00	0.00	0.00	95.72	

Field 1 - 11/01/2021: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	147.93	553,947.00 *gal*
Application event totals		0.00	0.00	0.00	147.93	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	182.73	684,287.00 *gal*
Application event totals		0.00	0.00	0.00	182.73	

Field 2 - 11/03/2022: Oats, hay

Field name: Field 2

Crop: Oats, hay

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	48.93	12.50	150.39	623.06	175,538.00 *gal*
IW	Ground water	6.74	0.00	0.00	373.42	1,188,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	55.11	619,117.00 *gal*
Application event totals		55.67	12.50	150.39	1,051.59	

11/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	48.62	12.42	149.45	619.16	174,440.00 *gal*
IW	Ground water	8.99	0.00	0.00	497.90	1,584,000.00 *gal*
Application event totals		57.61	12.42	149.45	1,117.05	

Field 2 - 11/03/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	28.40	7.26	87.30	361.67	101,897.00 *gal*
Canal	Surface water	0.00	0.00	0.00	58.75	660,000.00 *gal*
Application event totals		28.40	7.26	87.30	420.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	50.25	5.81	74.35	497.24	104,414.00 *gal*
Application event totals		50.25	5.81	74.35	497.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	10.49	0.00	0.00	580.88	1,848,000.00 *gal*
Application event totals		10.49	0.00	0.00	580.88	

Field 2 - 06/29/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	62.41	9.47	211.45	977.69	204,346.00 *gal*
WW Application-D1	Process wastewater	103.49	8.99	212.35	923.03	254,469.00 *gal*
Canal	Surface water	0.00	0.00	0.00	336.46	3,779,872.00 *gal*
Application event totals		165.90	18.46	423.81	2,237.18	

Field 2 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	223.34	2,509,053.00 *gal*
Application event totals		0.00	0.00	0.00	223.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	203.04	2,280,957.00 *gal*
Application event totals		0.00	0.00	0.00	203.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	211.74	2,378,712.00 *gal*
Application event totals		0.00	0.00	0.00	211.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	36.06	8.21	158.56	545.71	144,675.00 *gal*
Canal	Surface water	0.00	0.00	0.00	188.53	2,118,032.00 *gal*
Application event totals		36.06	8.21	158.56	734.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	8.99	0.00	0.00	497.90	1,584,000.00 *gal*
Application event totals		8.99	0.00	0.00	497.90	

Field 3 - 11/03/2022: Oats, hay

Field name: Field 3

Crop: Oats, hay

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 11/03/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	48.93	12.50	150.39	623.06	175,538.00 *gal*
IW	Ground water	6.74	0.00	0.00	373.42	1,188,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	55.11	619,117.00 *gal*
Application event totals		55.67	12.50	150.39	1,051.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	40.37	10.31	124.08	514.05	144,828.00 *gal*
IW	Ground water	8.99	0.00	0.00	497.90	1,584,000.00 *gal*
Application event totals		49.36	10.31	124.08	1,011.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	28.40	7.26	87.30	361.67	101,897.00 *gal*
Application event totals		28.40	7.26	87.30	361.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	50.25	5.81	74.35	497.24	104,414.00 *gal*
Application event totals		50.25	5.81	74.35	497.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	10.49	0.00	0.00	580.88	1,848,000.00 *gal*
Application event totals		10.49	0.00	0.00	580.88	

Field 3 - 06/29/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/29/2023

Field 3 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	103.49	8.99	212.35	923.03	254,469.00 *gal*
WW Application	Process wastewater	62.41	9.47	211.45	977.69	204,346.00 *gal*
Canal	Surface water	0.00	0.00	0.00	336.46	3,779,872.00 *gal*
Application event totals		165.90	18.46	423.81	2,237.18	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	223.34	2,509,053.00 *gal*
Application event totals		0.00	0.00	0.00	223.34	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	203.04	2,280,957.00 *gal*
Application event totals		0.00	0.00	0.00	203.04	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	211.74	2,378,712.00 *gal*
Application event totals		0.00	0.00	0.00	211.74	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	36.06	8.21	158.56	545.71	144,675.00 *gal*
Canal	Surface water	0.00	0.00	0.00	188.53	2,118,032.00 *gal*
Application event totals		36.06	8.21	158.56	734.24	

Field 3 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	8.99	0.00	0.00	497.90	1,584,000.00 *gal*
Application event totals		8.99	0.00	0.00	497.90	

Field 4 - 11/03/2022: Oats, hay

Field name: Field 4

Crop: Oats, hay

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	48.93	12.50	150.39	623.06	257,456.00 *gal*
IW	Ground water	5.11	0.00	0.00	282.90	1,320,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	75.15	1,238,234.00 *gal*
Application event totals		54.03	12.50	150.39	981.10	

12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	28.40	7.26	87.30	361.67	149,449.00 *gal*
Application event totals		28.40	7.26	87.30	361.67	

01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	43.79	5.06	64.80	433.33	133,457.00 *gal*
Application event totals		43.79	5.06	64.80	433.33	

Field 4 - 11/03/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	50.25	5.81	74.35	497.24	153,141.00 *gal*
Application event totals		50.25	5.81	74.35	497.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	8.94	0.00	0.00	495.07	2,310,000.00 *gal*
Application event totals		8.94	0.00	0.00	495.07	

Field 4 - 06/29/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	103.49	8.99	212.35	923.03	373,221.00 *gal*
WW Application	Process wastewater	62.41	9.47	211.45	977.69	299,708.00 *gal*
Canal	Surface water	0.00	0.00	0.00	348.06	5,734,978.00 *gal*
Application event totals		165.90	18.46	423.81	2,248.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	227.43	3,747,287.00 *gal*
Application event totals		0.00	0.00	0.00	227.43	

Field 4 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	213.58	3,519,191.00 *gal*
Application event totals		0.00	0.00	0.00	213.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	177.99	2,932,659.00 *gal*
Application event totals		0.00	0.00	0.00	177.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	36.06	8.21	158.56	545.71	212,190.00 *gal*
Canal	Surface water	0.00	0.00	0.00	138.43	2,280,957.00 *gal*
Application event totals		36.06	8.21	158.56	684.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	8.55	0.00	0.00	473.85	2,211,000.00 *gal*
Application event totals		8.55	0.00	0.00	473.85	

Field 5 - 12/05/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 12/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW	Ground water	10.14	0.00	0.00	561.86	4,290,000.00 *gal*
Application event totals		10.14	0.00	0.00	561.86	

Field 5 - 12/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	55.43	11.02	160.29	776.42	360,155.00 *gal*
Application event totals		55.43	11.02	160.29	776.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	104.78	9.10	215.00	934.52	618,328.00 *gal*
Canal	Surface water	0.00	0.00	0.00	130.52	3,519,191.00 *gal*
Application event totals		104.78	9.10	215.00	1,065.05	

Field 5 - 06/27/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	100.61	8.74	206.46	897.39	593,760.00 *gal*
Canal	Surface water	0.00	0.00	0.00	334.77	9,026,073.00 *gal*
Application event totals		100.61	8.74	206.46	1,232.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	216.33	5,832,733.00 *gal*
Application event totals		0.00	0.00	0.00	216.33	

Field 5 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	206.66	5,572,052.00 *gal*
Application event totals		0.00	0.00	0.00	206.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	55.56	5.99	114.55	657.28	509,975.00 *gal*
Canal	Surface water	0.00	0.00	0.00	169.20	4,561,914.00 *gal*
Application event totals		55.56	5.99	114.55	826.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	140.19	3,779,872.00 *gal*
Application event totals		0.00	0.00	0.00	140.19	

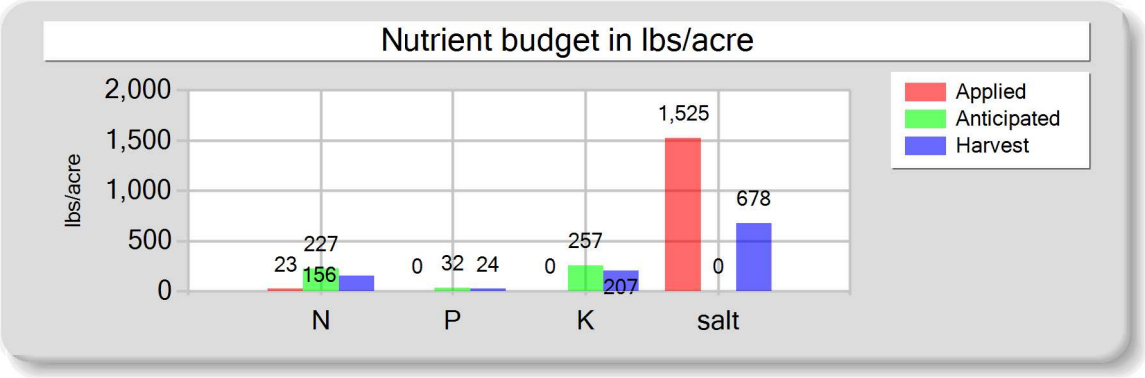
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	37.03	3.99	76.34	438.00	339,842.00 *gal*
Canal	Surface water	0.00	0.00	0.00	149.86	4,040,552.00 *gal*
Application event totals		37.03	3.99	76.34	587.86	

B. NUTRIENT BUDGET

Field 1 - 11/01/2021: Pasture

Field name: Field 1 Crop: Pasture Plant date: 11/01/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.99	0.00	0.00	1,524.68
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	22.99	0.00	0.00	1,524.68
Anticipated crop nutrient removal	226.50	31.50	256.50	0.00
Actual crop nutrient removal	156.00	24.00	207.00	677.82
Nutrient balance	-133.01	-24.00	-207.00	846.86
Applied to removed ratio	0.15	0.00	0.00	2.25

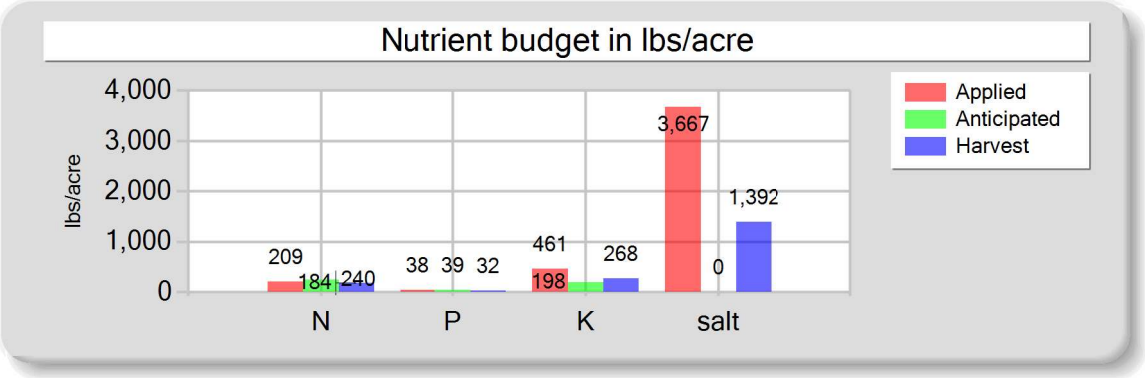
Fresh water applied	
4,373,042.00	*gallons*
161.04	*acre-inches*
32.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/03/2022: Oats, hay

Field name: Field 2 Crop: Oats, hay Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	176.19	37.99	461.50	2,101.13
Fresh water	26.22	0.00	0.00	1,566.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.41	37.99	461.50	3,667.19
Anticipated crop nutrient removal	240.00	39.00	198.00	0.00
Actual crop nutrient removal	184.00	31.72	268.12	1,391.76
Nutrient balance	25.41	6.27	193.38	2,275.42
Applied to removed ratio	1.14	1.20	1.72	2.63

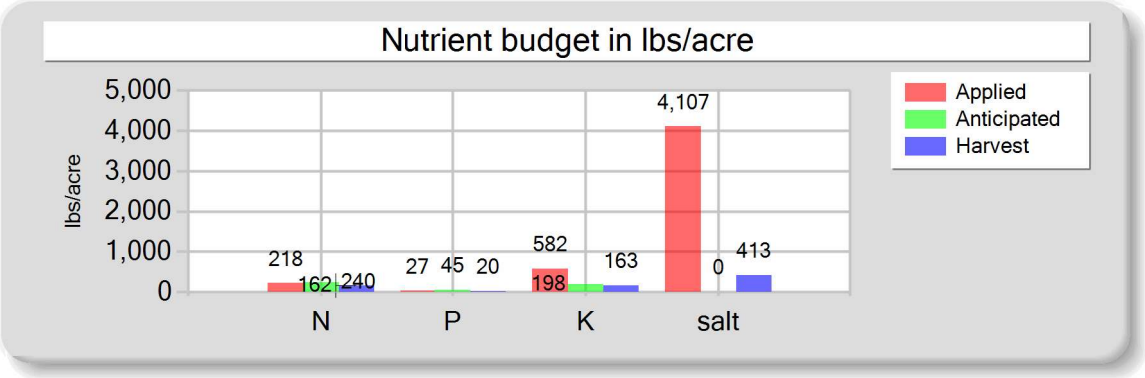
Fresh water applied	
5,899,117.00	*gallons*
217.24	*acre-inches*
14.48	*inches/acre*

Process wastewater applied	
556,289.00	*gallons*
20.49	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/29/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.96	26.67	582.37	2,446.43
Fresh water	8.99	0.00	0.00	1,661.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.95	26.67	582.37	4,107.43
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	162.48	19.71	163.22	412.62
Nutrient balance	55.47	6.96	419.15	3,694.80
Applied to removed ratio	1.34	1.35	3.57	9.95

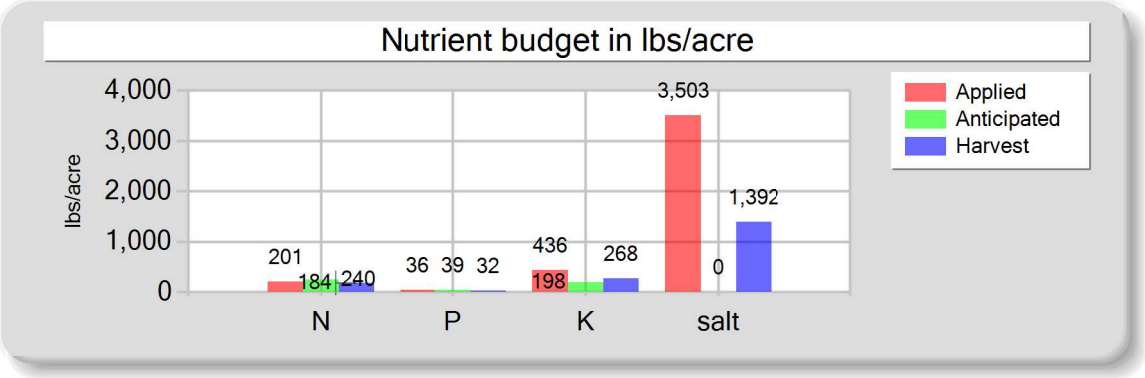
Fresh water applied	
14,650,626.00	*gallons*
539.53	*acre-inches*
35.97	*inches/acre*

Process wastewater applied	
603,490.00	*gallons*
22.22	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/03/2022: Oats, hay

Field name: Field 3 Crop: Oats, hay Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	167.94	35.88	436.13	1,996.02
Fresh water	26.22	0.00	0.00	1,507.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.16	35.88	436.13	3,503.33
Anticipated crop nutrient removal	240.00	39.00	198.00	0.00
Actual crop nutrient removal	184.00	31.72	268.12	1,391.76
Nutrient balance	17.16	4.16	168.01	2,111.57
Applied to removed ratio	1.09	1.13	1.63	2.52

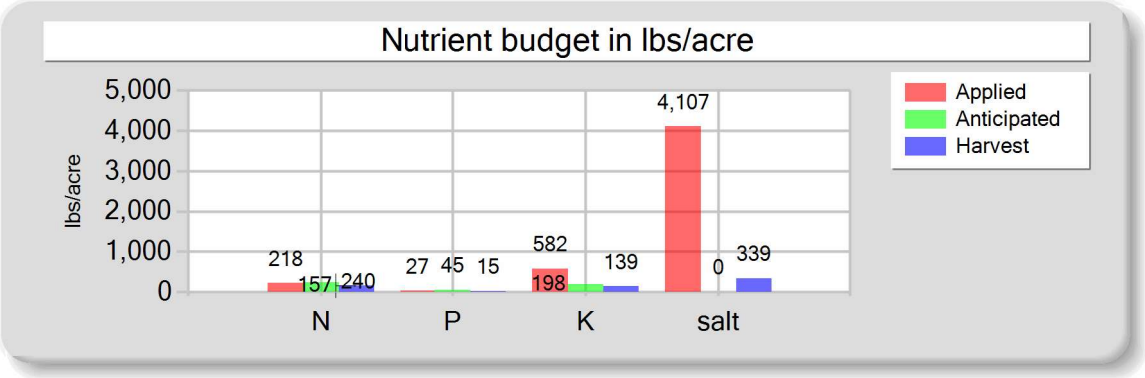
Fresh water applied	
5,239,117.00	*gallons*
192.94	*acre-inches*
12.86	*inches/acre*

Process wastewater applied	
526,677.00	*gallons*
19.40	*acre-inches*
1.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/29/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.96	26.67	582.37	2,446.43
Fresh water	8.99	0.00	0.00	1,661.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.95	26.67	582.37	4,107.43
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	157.27	14.50	139.43	339.30
Nutrient balance	60.68	12.17	442.94	3,768.12
Applied to removed ratio	1.39	1.84	4.18	12.11

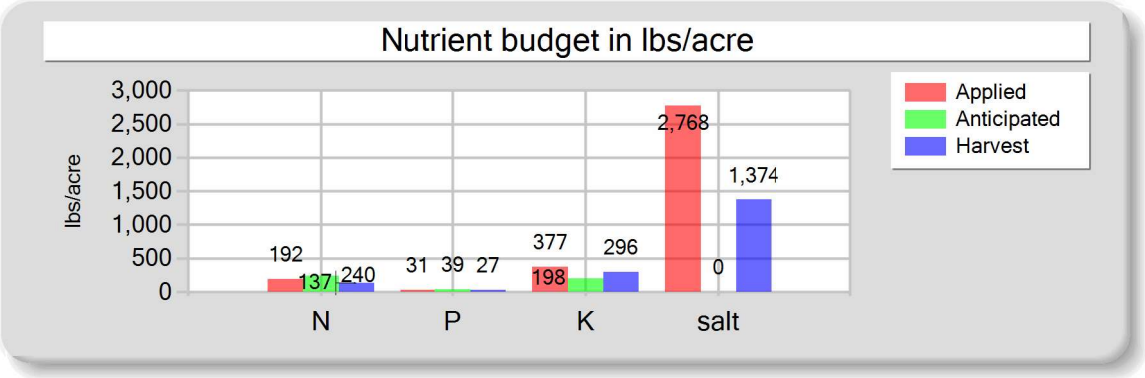
Fresh water applied	
14,650,626.00	*gallons*
539.53	*acre-inches*
35.97	*inches/acre*

Process wastewater applied	
603,490.00	*gallons*
22.22	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/03/2022: Oats, hay

Field name: Field 4 Crop: Oats, hay Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.36	30.63	376.84	1,915.30
Fresh water	14.04	0.00	0.00	853.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.41	30.63	376.84	2,768.42
Anticipated crop nutrient removal	240.00	39.00	198.00	0.00
Actual crop nutrient removal	137.21	26.99	296.15	1,374.15
Nutrient balance	55.20	3.64	80.70	1,394.27
Applied to removed ratio	1.40	1.13	1.27	2.01

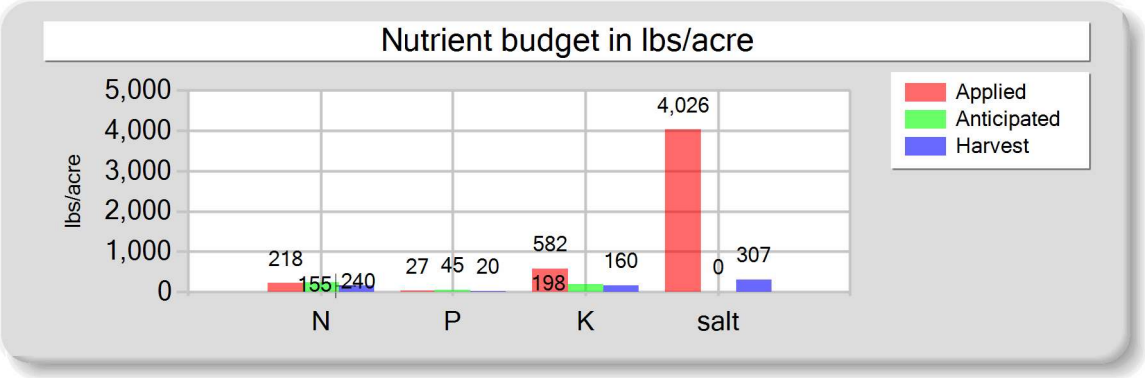
Fresh water applied	
4,868,234.00	*gallons*
179.28	*acre-inches*
8.15	*inches/acre*

Process wastewater applied	
693,503.00	*gallons*
25.54	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/29/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.96	26.67	582.37	2,446.43
Fresh water	8.55	0.00	0.00	1,579.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.51	26.67	582.37	4,025.77
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	154.67	20.08	160.25	306.74
Nutrient balance	62.84	6.59	422.12	3,719.03
Applied to removed ratio	1.41	1.33	3.63	13.12

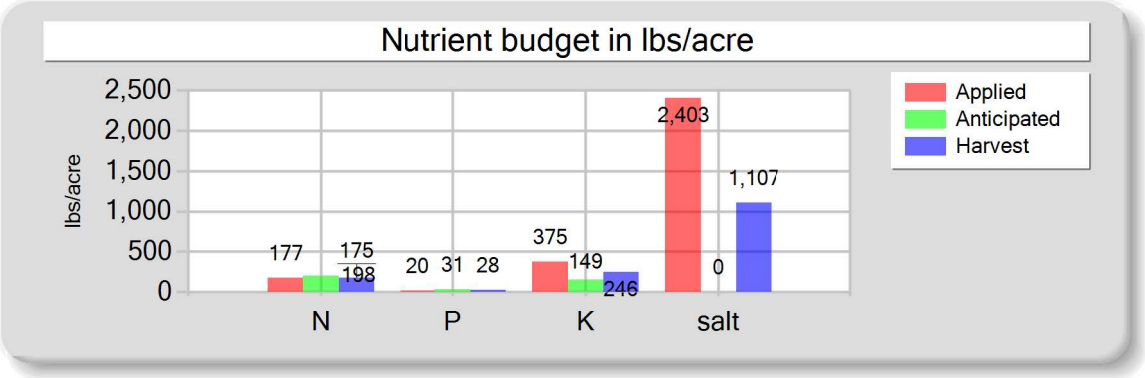
Fresh water applied	
20,426,072.00	*gallons*
752.22	*acre-inches*
34.19	*inches/acre*

Process wastewater applied	
885,119.00	*gallons*
32.60	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 12/05/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 12/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	160.21	20.12	375.29	1,710.94
Fresh water	10.14	0.00	0.00	692.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.35	20.12	375.29	2,403.33
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	174.54	27.52	246.48	1,106.70
Nutrient balance	2.81	-7.40	128.81	1,296.63
Applied to removed ratio	1.02	0.73	1.52	2.17

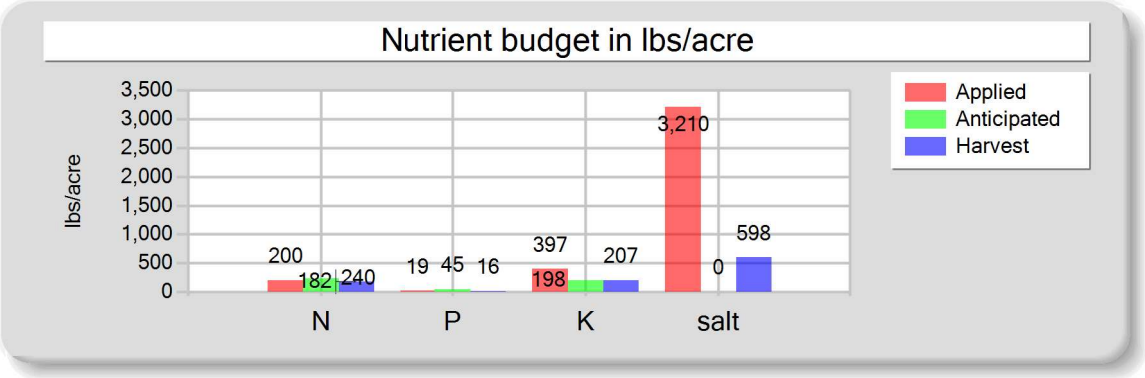
Fresh water applied	
7,809,191.00	*gallons*
287.59	*acre-inches*
7.99	*inches/acre*

Process wastewater applied	
978,483.00	*gallons*
36.03	*acre-inches*
1.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 06/27/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.20	18.73	397.34	1,992.67
Fresh water	0.00	0.00	0.00	1,217.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	200.20	18.73	397.34	3,209.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	182.18	16.27	206.76	598.02
Nutrient balance	18.02	2.46	190.58	2,611.65
Applied to removed ratio	1.10	1.15	1.92	5.37

Fresh water applied	
32,813,196.00	*gallons*
1,208.40	*acre-inches*
33.57	*inches/acre*

Process wastewater applied	
1,443,577.00	*gallons*
53.16	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

D2Spring SM

Sample and source description: D2Spring SM

Sample date: 05/09/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,830.00	2,460.00	13,600.00	8,840.00	7,090.00	1,580.00	2,000.00	595.00		52.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

D2 Fall SM

Sample and source description: D2 Fall SM

Sample date: 10/04/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	6,010.00	40,500.00							74.40
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	501.00	241.00			128.00	1,540.00								7,850.00	6,380
DL	0.50	0.50			0.50	0.50								1.00	10

1st QWW D1

Sample and source description: 1st QWW D1

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	664.00	314.00			132.00	1,920.00								10,000.00	9,300
DL	0.50	0.50			0.50	0.50								1.00	10

1st QWW D2

Sample and source description: 1st QWW D2

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	865.00	473.00			100.00	1,280.00								9,720.00	8,560
DL	0.50	0.50			0.50	0.50								1.00	10

D1 2nd Qtr WW

Sample and source description: D1 2nd Qtr WW

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	731.00	476.00			63.50	1,500.00								10,700.00	6,520
DL	0.50	0.50			0.50	0.50								1.00	10

D2 2nd QTR WW

Sample and source description: D2 2nd QTR WW

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	549.00	280.00			83.30	1,860.00								9,960.00	8,600
DL	0.50	0.50			0.50	0.50								1.00	10

3rd QWW D1

Sample and source description: 3rd QWW D1

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00	336.00			50.70	969.00								7,340.00	5,560
DL	0.50	0.50			0.50	0.50								1.00	10

3rd QWW D2

Sample and source description: 3rd QWW D2

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	448.00	241.00			102.00	1,970.00								9,980.00	6,780
DL	0.50	0.50			0.50	0.50								1.00	10

4th QWW-Dairy 2

Sample and source description: 4th QWW-Dairy 2

Sample date: 12/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	515.00	235.00			79.40	1,890.00	216.00	136.00	341.00	3,380.00	0.00	265.00	0.00	8,640.00	9,080
DL	0.50	0.50			0.50	0.50	0.50	0.50	0.50	0.50	2.00	0.50	0.80	1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

CCID

Sample description: CCID

Sample date: 05/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								200.00	160
DL	1.00		0.10								1.00	10

DW #1

Dairy 2 DW#1

Sample description: Dairy 2 DW#1

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	7.22								1,120.00	
DL		0.50	0.50								1.00	

DW #2

Dairy 2 DW#2

Sample description: Dairy 2 DW#2

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	7.99								1,160.00	
DL		0.50	0.50								1.00	

IW

IW

Dairy 2 IW

Sample description: Dairy 2 IW

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.20	0.00	7.44								1,090.00	565
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/01/2021: Pasture

Pasture

Sample and source description: Pasture

Sample date: 03/28/2024 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	6,900.00		7.90
DL	100.00	100.00	100.00		0.01

Field 2 - 11/03/2022: Oats, hay

Field 2 - 11/03/2022: Oats, hay

D2 Field 2 & 3

Sample and source description: D2 Field 2 & 3

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 5.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	1,810.00	15,300.00		8.44
DL	100.00	100.00	100.00		0.01

Field 2 - 06/29/2023: Corn, silage

D2 Field 2

Sample and source description: D2 Field 2

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,370.00	530.00	4,390.00		3.10
DL	100.00	100.00	100.00		0.01

Field 3 - 11/03/2022: Oats, hay

D2 Fields 2 & 3

Sample and source description: D2 Fields 2 & 3

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 5.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	1,810.00	15,300.00		8.44
DL	100.00	100.00	100.00		0.01

Field 3 - 06/29/2023: Corn, silage

D2 Field 3

Sample and source description: D2 Field 3

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,230.00	390.00	3,750.00		2.70
DL	100.00	100.00	100.00		0.01

Field 4 - 11/03/2022: Oats, hay

D2 Field 4

Sample and source description: D2 Field 4

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,830.00	1,540.00	16,900.00		8.36
DL	100.00	100.00	100.00		0.01

Field 4 - 06/29/2023: Corn, silage

D2 Field 4

Sample and source description: D2 Field 4

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,160.00	540.00	4,310.00		2.50
DL	100.00	100.00	100.00		0.01

Field 5 - 12/05/2022: Oats, silage-soft dough

Dairy 2

Sample and source description: Dairy 2

Sample date: 05/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,440.00	700.00	6,270.00		10.20
DL	100.00	100.00	100.00		0.01

Field 5 - 06/27/2023: Corn, silage

D2 Field 5

Sample and source description: D2 Field 5

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,040.00	450.00	5,720.00		4.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

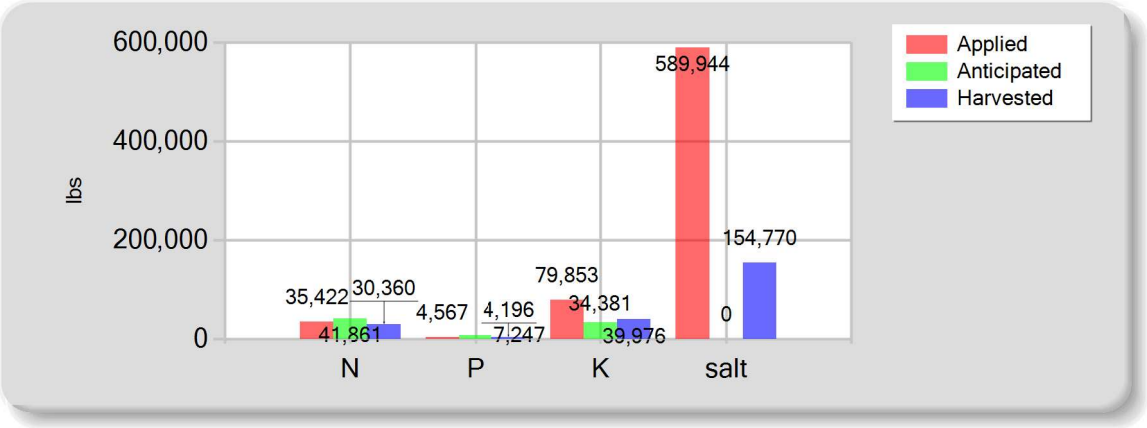
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

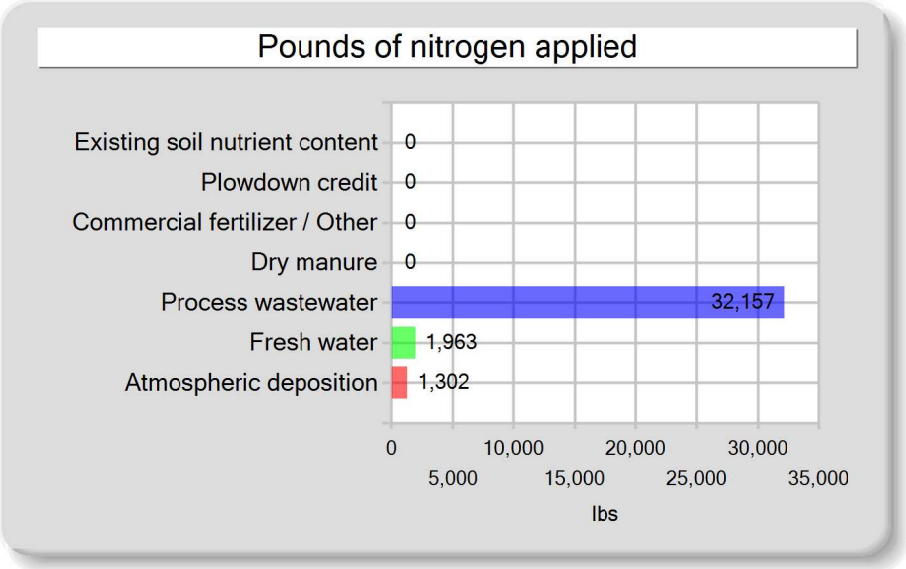
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

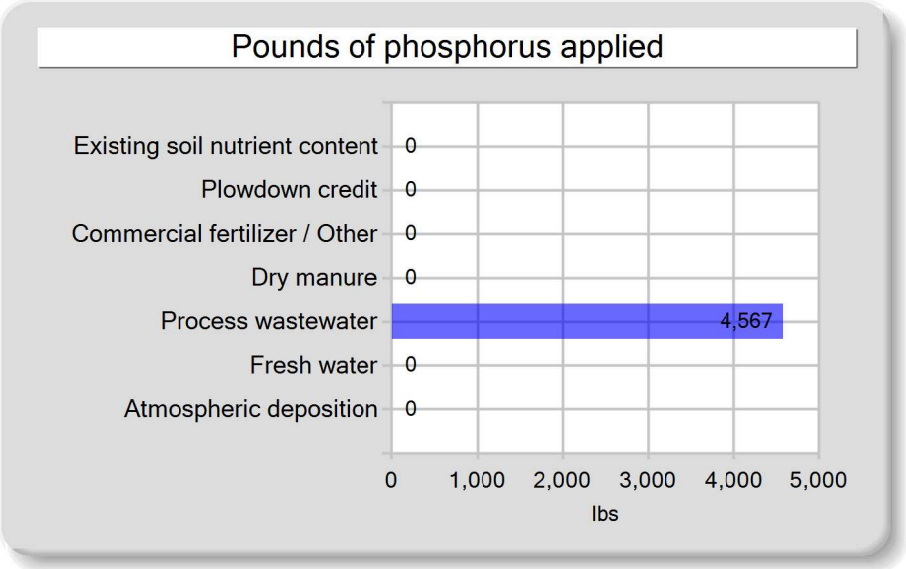
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	32,156.61	4,567.17	79,852.89	364,138.22
Fresh water	1,963.44	0.00	0.00	225,805.77
Atmospheric deposition	1,302.00	0.00	0.00	0.00
Total nutrients applied	35,422.05	4,567.17	79,852.89	589,943.99
Anticipated crop nutrient removal	41,860.50	7,247.10	34,380.90	0.00
Actual crop nutrient removal	30,359.74	4,196.28	39,975.76	154,770.31
Nutrient balance	5,062.31	370.89	39,877.13	435,173.68
Applied to removed ratio	1.17	1.09	2.00	3.81

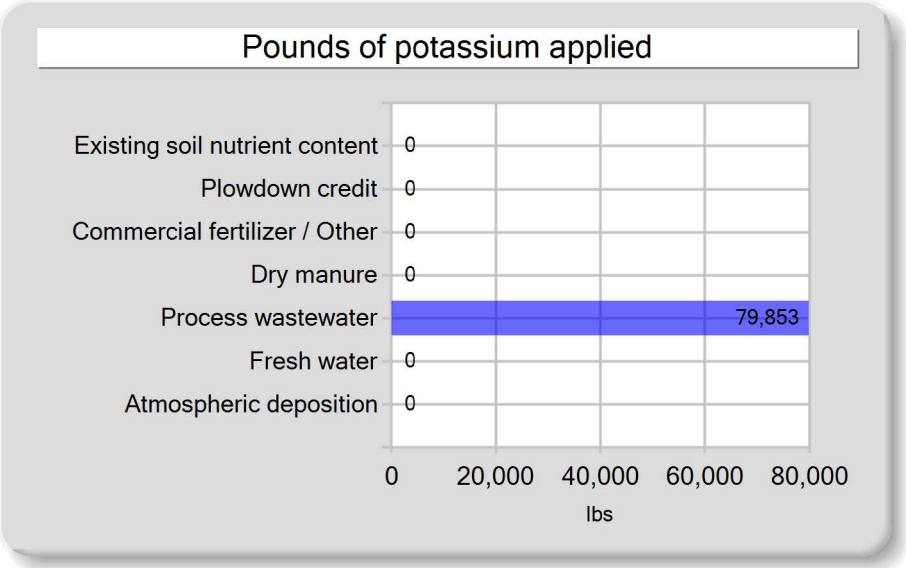
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

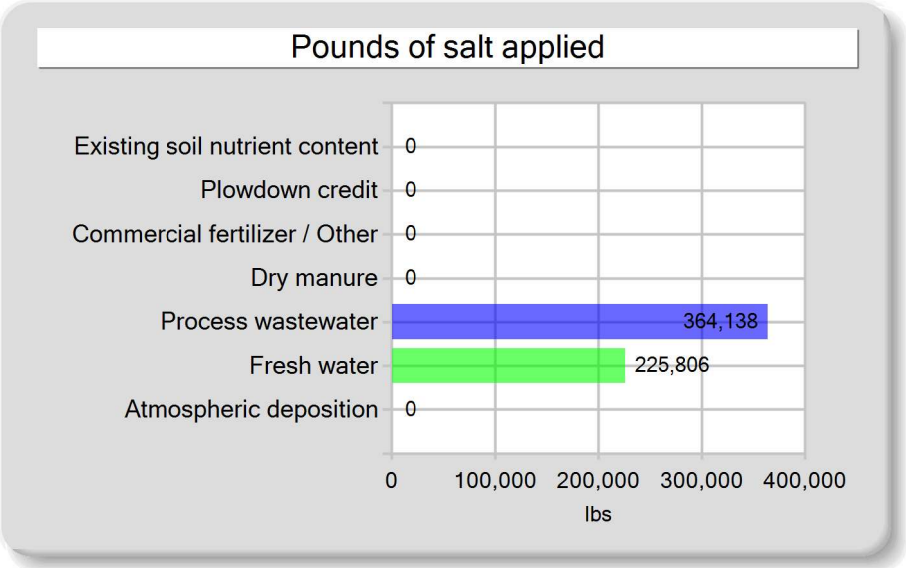


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Additional solid manure was used onsite and will be reported in the 2024 Annual Report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Pat Amarante (Jun 28, 2024 14:34 PDT)	Martins View Jersey Dairy, Inc. (Jun 29, 2024 17:26 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Pat Amarante	Martins View Jersey Dairy, Inc.
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/28/2024	06/29/2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Martins View Jersey Dairy
1000 Red Lion Ct.
Newman, CA 95360

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/14/23 7:30 AM
SUBMITTED BY: KRC
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 7.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy 2 DW#1	T3348003A	Electrical Conductivity (EC)	EPA 120.1 and	1.12	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.22	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	714	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	98.2 (T3346035A)	98.8 (T3346035A)	0.608
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Martins View Jersey Dairy
1000 Red Lion Ct.
Newman, CA 95360

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/14/23 7:30 AM
SUBMITTED BY: KRC
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 6.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy 2 DW#2	T3348003B	Electrical Conductivity (EC)	EPA 120.1 and	1.16	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.99	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	745	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	98.2 (T3346035A)	98.8 (T3346035A)	0.608
Electrical Conductivity (EC)	100			0.514
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

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Compliance Analysis Report

Customer
Martins View Jersey Dairy
1000 Red Lion Ct.
Newman, CA 95360

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/14/23 7:30 AM
SUBMITTED BY: KRC
ANALYZED DATE: 12/15/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 7.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy 2 IW	T3348003C	Electrical Conductivity (EC)	EPA 120.1 and	1.09	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.44	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	565	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	10.2	mg/L
		Soluble Salts (SALT-SOL)	Calculation	696	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	100	98.2 (T3346035A)	98.8 (T3346035A)			0.608
Electrical Conductivity (EC)	100			1.16 (T3348003B)	1.17 (Dup)	0.514
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)			4.70
Total Dissolved Solids (TDS)	101			880 (W3346003B)	860 (Dup)	2.30
Total Nitrogen (TN)	99.1			0.000 (T3345031B)	0.000 (null)	

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Martins New
Mailing address: 1000 Redhawk
(not facility address) Newman, CA 95360
Phone#:

Authorized Copy Released To: KR Cordero
Sampled By: A. Martins
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/12/23	7:25pm	Dairy 2 DW#1	DW	7.2°C		X				
2		7:26pm	Dairy 2, DW#2		6.8°C		X				
3		7:30pm	Dairy 2, IW	IW	7.2°C						H7
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]	KR Cordero		12/13/23		
2nd	[signature]	DENELE ANALYTICAL, INC.	12-14-23 7:30			T 33 48003

Required to be submitted to any agency? (YES / NO)

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Martins View Jersey Dairy, Inc.

Name of Dairy Facility: Martins View Jersey Dairy #2

Facility Address:

28336 W Bambauer RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Antonio Martins (209) 678-1156
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilzer

Address of Hauling Company/Person:

1275 North AVE	Gustine	CA	95360
Number and Street	City	State	Zip Code

Contact Person: John & Florie Nunes (209) 495-1964
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

J&F Fertilzer (209) 495-1964
Name Phone Number

1275 North AVE	Gustine	CA	95322
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

1275 North AVE	Gusstine	95322
Address	City	Zip Code

	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 12/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 48.00 tons

Manure Solids Content: 56.6 %

Method used to determine amount of manure:

Average Load Weight Multiplied by Total Loads.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Antonio Martins/Martins View Jersey Dairy, Inc. (Jul 1, 2024 12:54 PDT) — 07/01/2024

Operator Signature — Date

john nunes — 06/28/2024

john nunes (Jul 1, 2024 09:42 PDT)

Hauler Signature — Date